

Ex. No: 4

Roll No: 240701013

DATA MODELLING AND FORM DESIGN

AIM

To design a data model and create forms in Zoho Creator, define field types and validations, and establish relationships between forms using Lookup fields, by building a Library Management application.

SOFTWARE / TOOLS USED

- Zoho Creator (Low-Code Application Development Platform)
- Web browser (Google Chrome / Microsoft Edge)

THEORY

Data modelling is the process of identifying the entities of an application, the attributes of each entity and the relationships between them. In Zoho Creator, each entity is represented as a Form, each attribute as a Field, and the stored entries are called Records. Fields have types such as Single Line, Number, Date, Dropdown, Radio and Email, and can be given properties like mandatory, unique, default value and validation. Relationships between forms are created using the Lookup field, which links a record in one form to a record in another form (one-to-many relationship).

DATA MODEL

The application “Library Management” contains three forms:

Form	Field Name	Field Type	Description
Members	Member_Name	Name	Full name of the member
Members	Email	Email	Member email (unique)
Members	Phone	Phone Number	Contact number
Members	Member_Type	Dropdown	Student / Staff
Books	Book_Title	Single Line	Title of the book
Books	Author	Single Line	Author name
Books	Category	Dropdown	Subject / genre
Books	Copies_Available	Number	Number of copies in stock
Book_Issue	Member	Lookup (Members)	Links issue record to a member
Book_Issue	Book	Lookup (Books)	Links issue record to a book
Book_Issue	Issue_Date	Date	Date of issue (default: today)

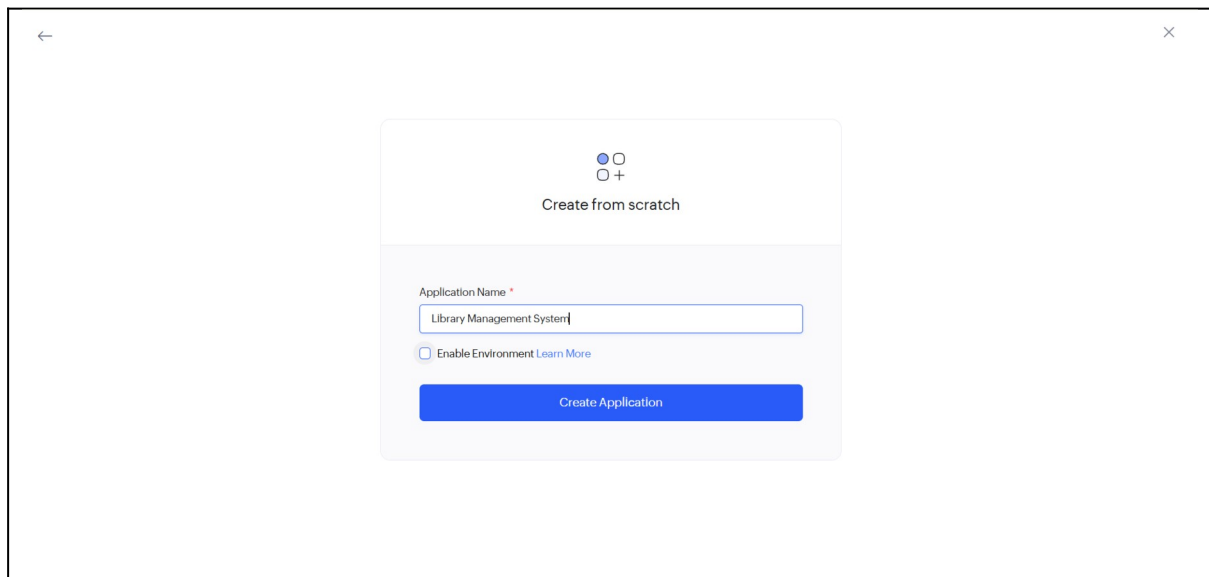
Form	Field Name	Field Type	Description
Book_Issue	Due_Date	Date	Last date for return
Book_Issue	Status	Radio	Issued / Returned

Relationship: One Member can have many Book_Issue records, and one Book can appear in many Book_Issue records (Book_Issue is the child form; Members and Books are the parent forms).

PROCEDURE

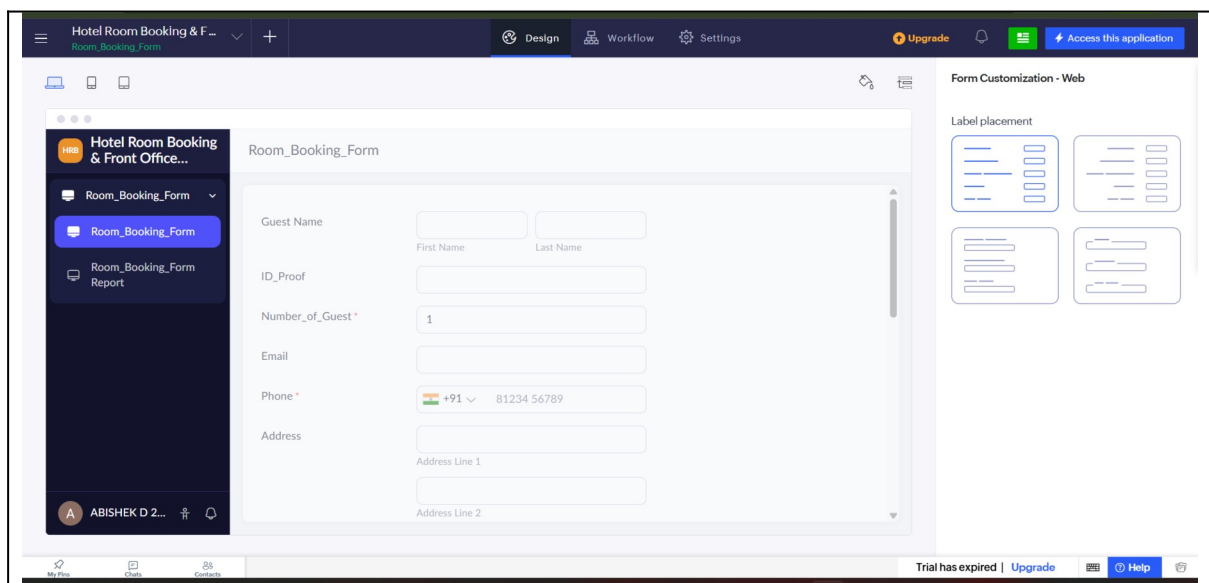
1. **Create the application:** Sign in to Zoho Creator, click Create Application → Create from scratch, enter the application name as “Library Management” and click Create.
2. **Create the Members form:** Click Create Form → Start from scratch, name the form “Members”, and drag the fields Name, Email, Phone Number and Dropdown (Member_Type) from the left pane onto the form canvas. Set the dropdown choices to Student and Staff.
3. **Create the Books form:** Create another form named “Books” with fields Book_Title, Author, Category (Dropdown) and Copies_Available (Number).
4. **Create the Book_Issue form with relationships:** Create a form named “Book_Issue”. Add two Lookup fields, Member (choose form Members, field Member_Name) and Book (choose form Books, field Book_Title). Also add Issue_Date, Due_Date (Date) and Status (Radio: Issued / Returned).
5. **Set field properties and validations:** Open the properties of each field and set Mandatory for the required fields, Do not allow duplicate values for Email, and a default value of today’s date for Issue_Date. Add field validation where required (e.g., Copies_Available must not be negative).
6. **Save and run the application:** Click Save, then Access Application to open the live mode of the application.
7. **Enter sample records:** Fill the Members and Books forms with sample data and submit. Then open the Book_Issue form, select a member and a book from the Lookup dropdowns, enter the dates and submit.
8. **Verify the output:** Open the report of each form and verify that the submitted records are displayed correctly.

Screenshot 1: Creating the new application



The screenshot shows a 'Create from scratch' dialog box in Zoho Creator. At the top, there are two icons: a circle with a dot and a square with a plus sign. Below them is the text 'Create from scratch'. The dialog has a light gray background. In the center, there is a section with a white background. It contains the label 'Application Name *' followed by a text input field containing 'Library Management System'. Below the input field is a checkbox labeled 'Enable Environment' with a link 'Learn More' next to it. At the bottom of this section is a blue button labeled 'Create Application'.

Screenshot 2: Members form in the form builder



The screenshot shows the Zoho Creator form builder interface. The top navigation bar includes 'Design', 'Workflow', and 'Settings' tabs. The main workspace displays a form titled 'Room_Booking_Form'. The form fields are: 'Guest Name' (split into 'First Name' and 'Last Name'), 'ID_Proof', 'Number_of_Guest *' (with a value of '1'), 'Email', 'Phone *' (with a country code dropdown set to '+91' and a value of '81234 56789'), and 'Address' (split into 'Address Line 1' and 'Address Line 2'). On the right side, there is a 'Form Customization - Web' panel with 'Label placement' options. The left sidebar shows a list of applications, with 'Room_Booking_Form' selected. The bottom status bar indicates 'Trial has expired' and provides links for 'Upgrade', 'Help', and a trash icon.

RESULT

Thus, the data model was designed and the forms Members, Books and Book_Issue were created in Zoho Creator with suitable field types, validations and Lookup relationships, and the sample records were stored and verified successfully.